

Complex Analysis Qualifying Exam Review

Fall 2026

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Chapter 1

Fall 2025

Problem 1.1. 5 Let $\{f_n\}$ be a sequence of holomorphic functions on the open unit disk

$$\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\},$$

and suppose that $f_n \rightarrow f$ uniformly on compact subsets of $\mathbb{D}$, where $f : \mathbb{D} \rightarrow \mathbb{C}$ is continuous. Prove that f is holomorphic on $\mathbb{D}$.

Proof. Morera's theorem. □

Problem 1.2. 6 Use contour integration to evaluate

$$\int_0^\infty \frac{x^2}{x^4 + 1} dx.$$

Proof. Use residue theorem for $z_1 = e^{\frac{\pi i}{4}}, z_2 = e^{\frac{3\pi i}{4}}$. □

Problem 1.3. 7 Let f be an analytic function on the open unit disk

$$\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\},$$

and suppose that

$$|f(z)| \geq \sqrt{|z|}$$

for all $z \in \mathbb{D}$. Show that

$$|f(z)| \geq 1 \quad \text{for all } z \in \mathbb{D}.$$

Proof. Maximum modulus principle. □

Problem 1.4. 8 Determine how many solutions

$$z^7 - 7z^4 + z^3 - 3z = 1$$

are in the annulus

$$1 < |z| < 2.$$

Proof. Rouché's theorem. □

Chapter 2

Cauchy's integral formula

Problem 2.1. Evaluate the following integral:

$$\int_0^\infty \frac{1 - \cos x}{x^2} dx$$

Proof. The answer is

$$\int_0^\infty \frac{1 - \cos x}{x^2} dx = \frac{\pi}{2}$$

We use the contour of a semicircle skipping $x = 0$. Let $f(z) = \frac{1 - e^{iz}}{z^2}$, then by Cauchy's integral formula, we have

$$\int_{\gamma_R} f(z) dz + \int_{\gamma_\varepsilon} f(z) dz + \int_{|z| > \varepsilon} f(z) dz = 0 \quad (1)$$

where

$$\lim_{\varepsilon \rightarrow 0} \int_{|z| > \varepsilon} f(z) dz = 2 \int_0^\infty \frac{1 - \cos x}{x^2} dx$$

Hence it suffices to compute the first two terms in (1). Let $z = Re^{i\theta}$, we see that

$$\begin{aligned} \left| \int_{\gamma_R} f(z) dz \right| &= \left| \int_0^\pi \frac{1 - e^{iRe^{i\theta}}}{R^2 e^{2i\theta}} iRe^{i\theta} d\theta \right| \\ &\leq \int_0^\pi \frac{2}{R} d\theta \rightarrow 0 \end{aligned}$$

as $R \rightarrow \infty$. Similarly, we can compute $\int_{\gamma_\varepsilon} f(z) dz$ by setting $z = \varepsilon e^{i\theta}$ and letting $\varepsilon \rightarrow 0$.

$$\begin{aligned} \int_{\gamma_\varepsilon} f(z) dz &= \int_\pi^0 \frac{1 - e^{i\varepsilon e^{i\theta}}}{\varepsilon^2 e^{2i\theta}} i\varepsilon e^{i\theta} d\theta \\ &= i \int_\pi^0 \frac{1 - e^{i\varepsilon e^{i\theta}}}{\varepsilon e^{i\theta}} d\theta \rightarrow -\pi \end{aligned}$$

as $\varepsilon \rightarrow 0$, since

$$\lim_{a \rightarrow 0} \frac{1 - e^{ia}}{a} = -i$$

□

Problem 2.2. Let U be a region. Suppose f is holomorphic in U , and a sequence $\{z_n\} \subset U$ converges to some $z \in U$. If $f(z_n) = 0$ for all n , show that $f \equiv 0$.

Proof. Since $z_n \rightarrow z$, we know $f(z) = \lim_{z_n \rightarrow z} f(z_n) = 0$. Assume f is not identically zero, then writing it as a power series, we have

$$f(w) = (w - z)^n g(w)$$

where $g(w) = a_n + a_{n+1}(w - z) + a_{n+2}(w - z)^2 + \dots$, for some $a_n \neq 0$. We know $g(z) = a_n \neq 0$, hence there exists a neighborhood V around z such that $g(w) \neq 0$ whenever $w \in V$. Hence, if $w \in V$, then $f(w) = 0$ iff $w = z$. This is a contradiction to $z_n \rightarrow z$ and $f(z_n) = 0$ for all n . $\square$



Warning 2.1. This shows that a nonzero holomorphic function has isolated zeros.

Problem 2.3 (Spring 2018, Q4). Let f be an entire function, suppose that for each z_0 , the power series expansion

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$$

has at least one coefficient $a_n = 0$. Show that f must be a polynomial.

Proof. It suffices to show that for some $k \in \mathbb{N}$, $f^{(k)} \equiv 0$ because

$$a_n = \frac{1}{n!} f^{(n)}(z_0)$$

We first observe that

$$\bigcup_{n \in \mathbb{N}} \{z : f^{(n)} = 0\} = \mathbb{C}$$

Then there must exist some $k \in \mathbb{N}$ such that

$$\{z : f^{(k)} = 0\} \text{ is uncountable}$$

Thus $f^{(k)}$ has uncountably many zeros, which implies $f \equiv 0$.

Lemma 2.1. Let f be a nonzero holomorphic function, then f cannot have uncountably many zeros.

Proof. Let $Z = \{z \in \mathbb{C} : f(z) = 0\}$, we know that Z contains no limit point. We can tile the complex plane using unit squares and suppose Z is uncountable, the intersection of Z with one of the tiles needs to be uncountable. This is a contradiction because this intersection is compact and discrete, hence must be finite. $\square$

$\square$

Problem 2.4 (Fall 2019, Q2). Let f be an entire function such that

$$\sup_{|z| > R} |f(z)| \leq AR^k + B$$

for positive constants A, B and all R large. Then show f is a polynomial with degree at most k .

Proof. This problem is a direct application of Cauchy's inequality. Writing $f(z)$ as a power series around 0, we have

$$f(z) = \sum_{n=0}^{\infty} a_n z^n$$

where

$$a_n = \frac{1}{n!} f^{(n)}(0)$$

It suffices to show that for any $m > k$, we have

$$f^{(m)}(0) = 0$$

Using Cauchy's inequality, we get

$$\begin{aligned} |f^{(m)}(0)| &\leq \frac{m! \sup_{|z|=R} |f(z)|}{R^m} \\ &\leq \frac{m!(AR^k + B)}{R^m} \rightarrow 0 \end{aligned}$$

as $R \rightarrow \infty$ because $m > k$. Hence we are done. $\square$

Problem 2.5 (Fall 2023, Q1). (a) Show that if an entire function f satisfies

$$|f(z)| \leq \sqrt{|z|}$$

for all $z \in \mathbb{C}$, then $f(z)$ is a constant function.

(b) Why isn't the function $z^{1/2}$ a counterexample for Part (a)?

Proof. (a) We write f as the Taylor series about 0,

$$f(z) = \sum_{n=0}^{\infty} a_n z^n$$

where

$$a_n = \frac{1}{n!} f^{(n)}(0)$$

Using Cauchy's inequality, for any R large, if $n \geq 1$,

$$|f^{(n)}(0)| \leq \frac{n! \sup_{|z|=R} |f(z)|}{R^n} \rightarrow 0$$

as $R \rightarrow \infty$. Thus $f(z) = a_0$, for some $a_0 \in \mathbb{C}$.

(b) The function is not entire. $\square$

Problem 2.6 (Fall 2018, Q4). Suppose f, g are both entire functions such that $|f(z)| \leq |g(z)|$ for all $z \in \mathbb{C}$. Show that there exists $c \in \mathbb{C}$ such that $f = cg$.

Proof. We will use Liouville's theorem. It suffices to show that $h = f/g$ continuously extends to an entire function. Suppose $g(z_0) = 0$ for some z_0 , then we know $f(z_0) = 0$, hence writing

$$f(z) = (z - z_0)^n s(z), \quad g(z) = (z - z_0)^m t(z)$$

where $s(z_0) \neq 0, t(z_0) \neq 0$. We must have $n \geq m$, suppose if not, i.e., $m > n$, then

$$|s(z)| \leq |z - z_0|^{m-n} |t(z)|$$

which is a contradiction by letting $z = z_0$. Thus z_0 is a removable singularity for $h = f/g$. This implies h can be extended to an entire function, as desired. $\square$

Chapter 3

Residue formula

Problem 3.1. Compute

$$\int_0^\infty \frac{1}{1+x^2} dx$$

Problem 3.2. Compute the following integral: fix $0 < a < 1$,

$$\int_{-\infty}^\infty \frac{e^{ax}}{1+e^x} dx$$

Proof. Use rectangles and the residue at πi . □

Problem 3.3. Assume that $f : \mathbb{C} \rightarrow \mathbb{C}$ is an entire function, not identically equal to 0, and let

$$Z = \{z \in \mathbb{C} : f(z) = 0\}.$$

Prove that if Z is unbounded, then f has an essential singularity at ∞ .

Chapter 4

Spring 2026

Problem 4.1. Let f be a holomorphic function on $\{z : |z| < 1\}$ that extends to a continuous function on $\{z : |z| \leq 1\}$ and maps the closed unit disk to itself. Furthermore, suppose

$$f(0) = f'(0) = \dots = f^{(n-1)}(0) = 0.$$

Show that:

(1)

$$|f(z)| \leq |z|^n \quad \text{for all } |z| \leq 1.$$

(2) If there is some $P \in D(0, 1) \setminus \{0\}$ such that

$$|f(P)| = |P|^n$$

or if

$$|f^{(n)}(0)| = n!,$$

then

$$f(z) = e^{i\theta} z^n.$$

Problem 4.2 (Also Fall 2023, Q2). Let $-1 < a < 0$. Compute

$$\int_0^\infty \frac{x^a}{1+x} dx.$$

Problem 4.3. Two parts.

(1) Let

$$f : U = \{z : 0 < |z| < 1\} \longrightarrow \mathbb{C}$$

be holomorphic and nonconstant. If there exists a sequence $\{z_n\} \subset U$ that converges to 0 and satisfies

$$f(z_n) = 0,$$

then prove that 0 is an essential singularity of f .

(2) Show that if f is an entire function that satisfies

$$\lim_{|z| \rightarrow \infty} f(z) = \infty,$$

then f is a polynomial.

Problem 4.4. Describe all injective holomorphic maps that map

$$\mathbb{C} \cup \{\infty\} \quad \text{to} \quad \mathbb{C} \cup \{\infty\}$$

and also map

$$\mathbb{R} \cup \{\infty\} \quad \text{to} \quad \mathbb{R} \cup \{\infty\}.$$